

NFR Screenshots

Team PUL5E

Create Staging Account

```
curl -i -X POST "$STAGING_BASE_URL/auth/register" \
-H "Content-Type: application/json" \
--data "$(node -e 'process.stdout.write(JSON.stringify({
email: process.env.NFR_TEST_EMAIL,
password: process.env.NFR_TEST_PASSWORD,
displayName: process.env.NFR_TEST_DISPLAY_NAME
})))"
```



```
Megans-MacBook:nfr-tests meganazmanov$ NFR_ACCOUNT_PREP_ACK=staging-disposable-account \
> node performance/prepare-test-account.js
{
  "requirement": "Representative staging dataset for performance testing",
  "executedAt": "2026-09-29T16:45:08.424Z",
  "mode": "prepare-and-verify",
  "environment": "staging",
  "baseUrl": "https://mealchemy-backend-staging-otygydpdv7a-ey.a.run.app",
  "targets": {
    "vaults": 4,
    "pantryItems": 30,
    "shoppingLists": 8,
    "shoppingListItems": 6,
    "recipes": 20,
    "recipeIngredients": 3,
    "recipeSteps": 3
  },
  "actual": {
    "vaults": 4,
    "pantryItems": 30,
    "nfrShoppingLists": 8,
    "minimumItemsPerNfrShoppingList": 6,
    "nfrRecipes": 20
  },
  "checks": [
    {
      "name": "profile is configured",
      "passed": true
    },
    {
      "name": "preferences are readable",
      "passed": true
    },
    {
      "name": "vault target is met",
      "passed": true
    },
    {
      "name": "pantry target is met",
      "passed": true
    },
    {
      "name": "shopping-list target is met",
      "passed": true
    },
    {
      "name": "shopping-list item target is met",
      "passed": true
    },
    {
      "name": "recipe target is met",
      "passed": true
    }
  ],
  "passed": true
}
```

Explanation: Registers the disposable staging account used by the NFR tests.

Result: HTTP 201 response received. Token output is redacted in the evidence screenshot.

30 second read-load smoke test

NFR_VUS=10 NFR_DURATION=30s \

k6 run nfr-tests/performance/non-recommendation-load.js

```
alt-svc: h3=":443"; ma=2592000,h3-29=":443"; ma=2592000
[token redacted]

M K6

execution: local
script: nfr-tests/performance/non-recommendation-load.js
output: -

scenarios: (100.00%) 1 scenario, 10 max VUs, 1m0s max duration (incl. graceful stop):
  * read_only_load: 10 looping VUs for 30s (gracefulStop: 30s)

INFO[0001] Starting staging read load with 10 VUs for 30s. source=console

THRESHOLDS
checks
  ✓ 'rate=0.99' rate=100.00%

http_req_duration(measured:true)
  ✓ 'p(95)<2000' p(95)=386.1ms

http_req_failed(measured:true)
  ✓ 'rate=0.01' rate=0.00%

TOTAL RESULTS
checks_total.....: 242      7.587743/s
checks_succeeded...: 100.00% 242 out of 242
checks_failed.....: 0.00%    0 out of 242

  ✓ test account login succeeds
  ✓ login response contains a token
  ✓ pantry returns HTTP 200
  ✓ shopping-lists returns HTTP 200
  ✓ vaults returns HTTP 200
  ✓ recipes returns HTTP 200

HTTP
http_req_duration.....: avg=265.34ms min=192.21ms med=233.64ms max=837.98ms p(90)=331.03ms p(95)=386.7ms
  { expected_response:true }...: avg=265.34ms min=192.21ms med=233.64ms max=837.98ms p(90)=331.03ms p(95)=386.7ms
  { measured:true }.....: avg=263.22ms min=192.21ms med=233.64ms max=837.98ms p(90)=330.95ms p(95)=386.1ms
http_req_failed.....: 0.00% 0 out of 242
  { measured:true }.....: 0.00% 0 out of 240
http_reqs.....: 241      7.556389/s

EXECUTION
iteration_duration.....: avg=1.27s min=1.19s med=1.23s max=2.04s p(90)=1.33s p(95)=1.38s
iterations.....: 240      7.525835/s
vus.....: 10 min=0 max=10
vus_max.....: 10 min=0 max=10

NETWORK
data_received.....: 290 kB 9.1 kB/s
data_sent.....: 41 kB 1.3 kB/s

running (0m31.9s), 00/10 VUs, 240 complete and 0 interrupted iterations
read_only_load ✓ [=====] 10 VUs 30s
Megans-MacBook:MealChemy meganaznanov$
```

Explanation: Runs a short staging read-load test against pantry, shopping-list, vault, and recipe endpoints.

Result: All 242 checks succeeded; p95 latency was about 386 ms and request failures were 0%.

Authorization matrix export

k6 run \

--summary-export=nfr-tests/results/authorization-matrix.json \

nfr-tests/security/authorization-matrix.js

Explanation: Checks that missing-token and invalid-token requests are rejected across protected endpoints.

Result: Authorization failures were 0; 154 checks succeeded and 0 failed.

```
features: null
Megan@MacBook:Mealchemy$ k6 run \
  --summary-export=nfr-tests/results/authorization-matrix.json \
  nfr-tests/security/authorization-matrix.js

Grafana
K6

execution: local
script: nfr-tests/security/authorization-matrix.js
output: -

scenarios: (100.00%) 1 scenario, 1 max VUs, 2m30s max duration (incl. graceful stop):
  * authorization_matrix: 1 iterations shared among 1 VUs (maxDuration: 2m0s, gracefulStop: 30s)

THRESHOLDS

authorization_failures
  ✓ 'count=0' count=0

checks
  ✓ 'rate=1' rate=100.00%

TOTAL RESULTS

checks_total.....: 154      3.140407/s
checks_succeeded...: 100.00% 154 out of 154
checks_failed.....: 0.00%   0 out of 154

✓ test account login succeeds
✓ login response contains a token
✓ missing-token: vault members is rejected
✓ invalid-token: vault members is rejected
✓ missing-token: create vault member is rejected
✓ invalid-token: create vault member is rejected
✓ missing-token: delete vault member is rejected
✓ invalid-token: delete vault member is rejected
✓ missing-token: dietary restrictions is rejected
✓ invalid-token: dietary restrictions is rejected
✓ missing-token: user profile is rejected
✓ invalid-token: user profile is rejected
✓ missing-token: update user profile is rejected
✓ invalid-token: update user profile is rejected
✓ missing-token: logout is rejected
✓ invalid-token: logout is rejected
✓ missing-token: owned vaults is rejected
✓ invalid-token: owned vaults is rejected
✓ missing-token: vault by id is rejected
✓ invalid-token: vault by id is rejected
✓ missing-token: accessible vaults is rejected
✓ invalid-token: accessible vaults is rejected
✓ missing-token: create vault is rejected
✓ invalid-token: create vault is rejected
✓ missing-token: update vault is rejected
✓ invalid-token: update vault is rejected
✓ missing-token: delete vault is rejected
✓ invalid-token: delete vault is rejected
✓ missing-token: folder recipes is rejected
✓ invalid-token: folder recipes is rejected
✓ missing-token: recipe folders is rejected
✓ invalid-token: recipe folders is rejected
✓ missing-token: recipe folder link is rejected
✓ invalid-token: recipe folder link is rejected
✓ missing-token: create recipe folder link is rejected
✓ invalid-token: create recipe folder link is rejected
✓ missing-token: update recipe folder link is rejected
✓ invalid-token: update recipe folder link is rejected
✓ missing-token: delete recipe folder link is rejected
✓ invalid-token: delete recipe folder link is rejected
✓ missing-token: user preferences is rejected
✓ invalid-token: user preferences is rejected
✓ missing-token: update user preferences is rejected
✓ invalid-token: update user preferences is rejected
✓ missing-token: private vault folders is rejected
✓ invalid-token: private vault folders is rejected
✓ missing-token: vault folders is rejected
✓ invalid-token: vault folders is rejected
✓ missing-token: folder by name is rejected
✓ invalid-token: folder by name is rejected
✓ missing-token: folder by id is rejected
✓ invalid-token: folder by id is rejected
✓ missing-token: create folder is rejected
✓ invalid-token: create folder is rejected
✓ missing-token: update folder is rejected
✓ invalid-token: update folder is rejected
✓ missing-token: delete folder is rejected
✓ invalid-token: delete folder is rejected
✓ missing-token: ingredient categories is rejected
✓ invalid-token: ingredient categories is rejected
✓ missing-token: allergy options is rejected
✓ invalid-token: allergy options is rejected
✓ missing-token: ingredient catalogue is rejected
✓ invalid-token: ingredient catalogue is rejected
✓ missing-token: ingredient catalogue search is rejected
✓ invalid-token: ingredient catalogue search is rejected
✓ missing-token: add external ingredient is rejected
✓ invalid-token: add external ingredient is rejected
✓ missing-token: shopping lists is rejected
✓ invalid-token: shopping lists is rejected
✓ missing-token: create shopping list is rejected
✓ invalid-token: create shopping list is rejected
✓ missing-token: update shopping list is rejected
✓ invalid-token: update shopping list is rejected
✓ missing-token: delete shopping list is rejected
✓ invalid-token: delete shopping list is rejected
✓ missing-token: shopping list from recipe is rejected
✓ invalid-token: shopping list from recipe is rejected
✓ missing-token: add recipe to shopping list is rejected
✓ invalid-token: add recipe to shopping list is rejected
✓ missing-token: shopping list by id is rejected
```

TOTAL RESULTS

checks_total.....: 154 3.140407/s
checks_succeeded...: 100.00% 154 out of 154
checks_failed.....: 0.00% 0 out of 154

- ✓ test account login succeeds
- ✓ login response contains a token
- ✓ missing-token: vault members is rejected
- ✓ invalid-token: vault members is rejected
- ✓ missing-token: create vault member is rejected
- ✓ invalid-token: create vault member is rejected
- ✓ missing-token: delete vault member is rejected
- ✓ invalid-token: delete vault member is rejected
- ✓ missing-token: dietary restrictions is rejected
- ✓ invalid-token: dietary restrictions is rejected
- ✓ missing-token: user profile is rejected
- ✓ invalid-token: user profile is rejected
- ✓ missing-token: update user profile is rejected
- ✓ invalid-token: update user profile is rejected
- ✓ missing-token: logout is rejected
- ✓ invalid-token: logout is rejected
- ✓ missing-token: owned vaults is rejected
- ✓ invalid-token: owned vaults is rejected
- ✓ missing-token: vault by id is rejected
- ✓ invalid-token: vault by id is rejected
- ✓ missing-token: accessible vaults is rejected
- ✓ invalid-token: accessible vaults is rejected
- ✓ missing-token: create vault is rejected
- ✓ invalid-token: create vault is rejected
- ✓ missing-token: update vault is rejected
- ✓ invalid-token: update vault is rejected
- ✓ missing-token: delete vault is rejected
- ✓ invalid-token: delete vault is rejected
- ✓ missing-token: folder recipes is rejected
- ✓ invalid-token: folder recipes is rejected
- ✓ missing-token: recipe folders is rejected
- ✓ invalid-token: recipe folders is rejected
- ✓ missing-token: recipe folder link is rejected
- ✓ invalid-token: recipe folder link is rejected
- ✓ missing-token: create recipe folder link is rejected
- ✓ invalid-token: create recipe folder link is rejected
- ✓ missing-token: update recipe folder link is rejected
- ✓ invalid-token: update recipe folder link is rejected
- ✓ missing-token: delete recipe folder link is rejected
- ✓ invalid-token: delete recipe folder link is rejected
- ✓ missing-token: user preferences is rejected
- ✓ invalid-token: user preferences is rejected
- ✓ missing-token: update user preferences is rejected
- ✓ invalid-token: update user preferences is rejected
- ✓ missing-token: private vault folders is rejected
- ✓ invalid-token: private vault folders is rejected
- ✓ missing-token: vault folders is rejected
- ✓ invalid-token: vault folders is rejected
- ✓ missing-token: folder by name is rejected
- ✓ invalid-token: folder by name is rejected
- ✓ missing-token: folder by id is rejected
- ✓ invalid-token: folder by id is rejected
- ✓ missing-token: create folder is rejected
- ✓ invalid-token: create folder is rejected
- ✓ missing-token: update folder is rejected
- ✓ invalid-token: update folder is rejected
- ✓ missing-token: delete folder is rejected
- ✓ invalid-token: delete folder is rejected
- ✓ missing-token: ingredient categories is rejected
- ✓ invalid-token: ingredient categories is rejected
- ✓ missing-token: allergy options is rejected
- ✓ invalid-token: allergy options is rejected
- ✓ missing-token: ingredient catalogue is rejected
- ✓ invalid-token: ingredient catalogue is rejected
- ✓ missing-token: ingredient catalogue search is rejected
- ✓ invalid-token: ingredient catalogue search is rejected
- ✓ missing-token: add external ingredient is rejected
- ✓ invalid-token: add external ingredient is rejected
- ✓ missing-token: shopping lists is rejected
- ✓ invalid-token: shopping lists is rejected
- ✓ missing-token: create shopping list is rejected
- ✓ invalid-token: create shopping list is rejected
- ✓ missing-token: update shopping list is rejected
- ✓ invalid-token: update shopping list is rejected
- ✓ missing-token: delete shopping list is rejected
- ✓ invalid-token: delete shopping list is rejected
- ✓ missing-token: shopping list from recipe is rejected
- ✓ invalid-token: shopping list from recipe is rejected
- ✓ missing-token: add recipe to shopping list is rejected
- ✓ invalid-token: add recipe to shopping list is rejected
- ✓ missing-token: shopping list by id is rejected
- ✓ invalid-token: shopping list by id is rejected
- ✓ missing-token: create shopping item is rejected
- ✓ invalid-token: create shopping item is rejected
- ✓ missing-token: update shopping item is rejected
- ✓ invalid-token: update shopping item is rejected
- ✓ missing-token: toggle purchased item is rejected
- ✓ invalid-token: toggle purchased item is rejected
- ✓ missing-token: delete shopping item is rejected
- ✓ invalid-token: delete shopping item is rejected
- ✓ missing-token: batch delete shopping items is rejected
- ✓ invalid-token: batch delete shopping items is rejected
- ✓ missing-token: select all shopping items is rejected
- ✓ invalid-token: select all shopping items is rejected
- ✓ missing-token: deselect all shopping items is rejected
- ✓ invalid-token: deselect all shopping items is rejected
- ✓ missing-token: complete shopping list is rejected

```

✓ invalid-token: ingredient categories is rejected
✓ missing-token: allergy options is rejected
✓ invalid-token: allergy options is rejected
✓ missing-token: ingredient catalogue is rejected
✓ invalid-token: ingredient catalogue is rejected
✓ missing-token: ingredient catalogue search is rejected
✓ invalid-token: ingredient catalogue search is rejected
✓ missing-token: add external ingredient is rejected
✓ invalid-token: add external ingredient is rejected
✓ missing-token: shopping lists is rejected
✓ invalid-token: shopping lists is rejected
✓ missing-token: create shopping list is rejected
✓ invalid-token: create shopping list is rejected
✓ missing-token: update shopping list is rejected
✓ invalid-token: update shopping list is rejected
✓ missing-token: delete shopping list is rejected
✓ invalid-token: delete shopping list is rejected
✓ missing-token: shopping list from recipe is rejected
✓ invalid-token: shopping list from recipe is rejected
✓ missing-token: add recipe to shopping list is rejected
✓ invalid-token: add recipe to shopping list is rejected
✓ missing-token: shopping list by id is rejected
✓ invalid-token: shopping list by id is rejected
✓ missing-token: create shopping item is rejected
✓ invalid-token: create shopping item is rejected
✓ missing-token: update shopping item is rejected
✓ invalid-token: update shopping item is rejected
✓ missing-token: toggle purchased item is rejected
✓ invalid-token: toggle purchased item is rejected
✓ missing-token: delete shopping item is rejected
✓ invalid-token: delete shopping item is rejected
✓ missing-token: batch delete shopping items is rejected
✓ invalid-token: batch delete shopping items is rejected
✓ missing-token: select all shopping items is rejected
✓ invalid-token: select all shopping items is rejected
✓ missing-token: deselect all shopping items is rejected
✓ invalid-token: deselect all shopping items is rejected
✓ missing-token: complete shopping list is rejected
✓ invalid-token: complete shopping list is rejected
✓ missing-token: units of measurement is rejected
✓ invalid-token: units of measurement is rejected
✓ missing-token: nutritional goals is rejected
✓ invalid-token: nutritional goals is rejected
✓ missing-token: recipe ingredients is rejected
✓ invalid-token: recipe ingredients is rejected
✓ missing-token: create recipe ingredient is rejected
✓ invalid-token: create recipe ingredient is rejected
✓ missing-token: update recipe ingredient is rejected
✓ invalid-token: update recipe ingredient is rejected
✓ missing-token: delete recipe ingredient is rejected
✓ invalid-token: delete recipe ingredient is rejected
✓ missing-token: pantry is rejected
✓ invalid-token: pantry is rejected
✓ missing-token: create pantry item is rejected
✓ invalid-token: create pantry item is rejected
✓ missing-token: update pantry item is rejected
✓ invalid-token: update pantry item is rejected
✓ missing-token: delete pantry item is rejected
✓ invalid-token: delete pantry item is rejected
✓ missing-token: pantry search is rejected
✓ invalid-token: pantry search is rejected
✓ missing-token: recipes is rejected
✓ invalid-token: recipes is rejected
✓ missing-token: community recipes is rejected
✓ invalid-token: community recipes is rejected
✓ missing-token: recipe by id is rejected
✓ invalid-token: recipe by id is rejected
✓ missing-token: create recipe is rejected
✓ invalid-token: create recipe is rejected
✓ missing-token: copy recipe is rejected
✓ invalid-token: copy recipe is rejected
✓ missing-token: recipe photo upload URL is rejected
✓ invalid-token: recipe photo upload URL is rejected
✓ missing-token: update recipe is rejected
✓ invalid-token: update recipe is rejected
✓ missing-token: delete recipe is rejected
✓ invalid-token: delete recipe is rejected
✓ missing-token: recipe steps is rejected
✓ invalid-token: recipe steps is rejected
✓ missing-token: create recipe step is rejected
✓ invalid-token: create recipe step is rejected
✓ missing-token: update recipe step is rejected
✓ invalid-token: update recipe step is rejected
✓ missing-token: reorder recipe steps is rejected
✓ invalid-token: reorder recipe steps is rejected
✓ missing-token: delete recipe step is rejected
✓ invalid-token: delete recipe step is rejected
✓ missing-token: flavour profile options is rejected
✓ invalid-token: flavour profile options is rejected
✓ missing-token: equipment is rejected
✓ invalid-token: equipment is rejected
✓ valid token: owned vaults returns HTTP 200
✓ valid token: pantry returns HTTP 200
✓ valid token: shopping lists returns HTTP 200
✓ valid token: recipes returns HTTP 200

```

```

CUSTOM
authorization_failures.....: 0      0/s

```

```

HTTP
http_req_duration.....: avg=314.9ms min=179.45ms med=187.7ms max=17.63s p(90)=263.85ms p(95)=271.8ms
{ expected_response:true }...: avg=3.74s min=230.41ms med=275.19ms max=17.63s p(90)=10.71s p(95)=14.17s
http_req_failed.....: 96.73% 148 out of 153
http_reqs.....: 153      3.120015/s

```

```

EXECUTION
iteration_duration.....: avg=30.95s min=30.95s med=30.95s max=30.95s p(90)=30.95s p(95)=30.95s
iterations.....: 1      0.020392/s
vus.....: 1      min=0      max=1
vus_max.....: 1      min=1      max=1

```

```

NETWORK
data_received.....: 37 kB 763 B/s
data_sent.....: 20 kB 408 B/s

```

```

running (0m49.0s), 0/1 VUs, 1 complete and 0 interrupted iterations
authorization_matrix / [=====] 1 VUs 0m31.0s/2m0s 1/1 shared iters
Megan's MacBook:MealChemy MeganZamanov

```

Feature:beef.FunctionalRequirementsMealChemy 0.0.4 91.0 6 Like Share 1 Just Ready

Performance baseline - 10 virtual users

```
NFR_VUS=10 \  
NFR_DURATION=5m \  
NFR_TEST_PROFILE=performance \  
NFR_SUMMARY_PATH=results/performance-baseline-10vu.json \  
k6 run performance/non-recommendation-load.js
```

Explanation: Runs the 5 minute baseline read-load scenario with 10 virtual users.

Result: 1841 checks passed, 0 failed; HTTP failure rate was 0 and p95 latency was 594.647 ms.

```
● Megan's-MacBook:nfr-tests meganazmanov$ NFR_VUS=10 \  
> NFR_DURATION=5m \  
> NFR_TEST_PROFILE=performance \  
> NFR_SUMMARY_PATH=results/performance-baseline-10vu.json \  
> k6 run performance/non-recommendation-load.js
```



```
execution: local  
script: performance/non-recommendation-load.js  
output: -  
  
scenarios: (100.00%) 1 scenario, 10 max VUs, 5m30s max duration (incl. graceful stop):  
  * read_only_load: 10 looping VUs for 5m0s (gracefulStop: 30s)  
  
INFO[0001] Starting staging performance read load with 10 VUs for 5m. source=console  
NFR SAFE SUMMARY  
test profile: performance  
environment: staging  
virtual users: 10  
duration: 5m  
checks passed: 1841  
checks failed: 0  
HTTP requests: 1840  
HTTP request rate: 5.684/s  
p95 duration: 594.647 ms  
HTTP failure rate: 0  
vaults p95: 363.748 ms; failure rate: 0  
pantry p95: 347.242 ms; failure rate: 0  
shopping-lists p95: 543.447 ms; failure rate: 0  
recipes p95: 960.397 ms; failure rate: 0  
sanitized evidence: results/performance-baseline-10vu.json  
  
running (5m23.7s), 00/10 VUs, 1839 complete and 0 interrupted iterations  
read only load ✓ [=====] 10 VUs 5m0s
```

Performance peak - 20 virtual users

```
NFR_VUS=20 \  
NFR_DURATION=5m \  
NFR_TEST_PROFILE=performance \  
NFR_SUMMARY_PATH=results/performance-peak-20vu.json \  
k6 run performance/non-recommendation-load.js
```

Explanation: Runs the 5 minute peak read-load scenario with 20 virtual users.

Result: 3963 checks passed, 0 failed; HTTP failure rate was 0 and p95 latency was 816.826 ms.

```
● Megans-MacBook:nfr-tests meganazmanov$ NFR_VUS=20 \  
> NFR_DURATION=5m \  
> NFR_TEST_PROFILE=performance \  
> NFR_SUMMARY_PATH=results/performance-peak-20vu.json \  
> k6 run performance/non-recommendation-load.js
```



```
execution: local  
script: performance/non-recommendation-load.js  
output: -  
  
scenarios: (100.00%) 1 scenario, 20 max VUs, 5m30s max duration (incl. graceful stop):  
  * read_only_load: 20 looping VUs for 5m0s (gracefulStop: 30s)  
  
INFO[0002] Starting staging performance read load with 20 VUs for 5m. source=console  
NFR SAFE SUMMARY  
test profile: performance  
environment: staging  
virtual users: 20  
duration: 5m  
checks passed: 3963  
checks failed: 0  
HTTP requests: 3962  
HTTP request rate: 13.053/s  
p95 duration: 816.826 ms  
HTTP failure rate: 0  
vaults p95: 383.466 ms; failure rate: 0  
pantry p95: 411.882 ms; failure rate: 0  
shopping-lists p95: 728.549 ms; failure rate: 0  
recipes p95: 1308.145 ms; failure rate: 0  
sanitized evidence: results/performance-peak-20vu.json  
  
running (5m03.5s), 00/20 VUs, 3961 complete and 0 interrupted iterations  
read_only_load ✓ [=====] 20 VUs 5m0s
```

Compare baseline and peak

```
node performance/compare-results.js \  
results/performance-baseline-10vu.json \  
results/performance-peak-20vu.json
```

Explanation: Shows the completed comparison checks for the baseline and peak performance results.

Result: All comparison checks passed, including p95 threshold, error-rate threshold, and throughput threshold.

```
● Megans-MacBook:nfr-tests meganazmanov$ node performance/compare-results.js \
> results/performance-baseline-10vu.json \
> results/performance-peak-20vu.json
{
  "baseline": {
    "path": "results/performance-baseline-10vu.json",
    "environment": "staging",
    "testProfile": "performance",
    "virtualUsers": 10,
    "duration": "5m",
    "p95Milliseconds": 594.647,
    "errorRate": 0,
    "requestsPerSecond": 5.684
  },
  "peak": {
    "path": "results/performance-peak-20vu.json",
    "environment": "staging",
    "testProfile": "performance",
    "virtualUsers": 20,
    "duration": "5m",
    "p95Milliseconds": 816.826,
    "errorRate": 0,
    "requestsPerSecond": 13.053
  },
  "latencyIncreasePercent": 37.363,
  "throughputRatio": 2.296,
  "checks": [
    {
      "name": "baseline is staging performance profile with 10 VUs for 5m",
      "passed": true
    },
    {
      "name": "peak is staging performance profile with 20 VUs for 5m",
      "passed": true
    },
    {
      "name": "baseline p95 <= 2000 ms",
      "passed": true
    },
    {
      "name": "peak p95 <= 2000 ms",
      "passed": true
    },
    {
      "name": "baseline error rate < 1%",
      "passed": true
    },
    {
      "name": "peak error rate < 1%",
      "passed": true
    },
    {
      "name": "peak p95 increase <= 50%",
      "passed": true
    },
    {
      "name": "peak throughput >= 1.8x baseline",
      "passed": true
    }
  ],
  "passed": true
}
```

Resilience read-load test - 100 virtual users

NFR_VUS=100 \

NFR_DURATION=5m \

Explanation: Runs a heavier 5 minute staging read-load scenario at 100 virtual users.
Result: 15890 checks passed and 1 failed; HTTP failure rate remained 0, with p95 latency at 2127.841 ms.

Code evidence can be found in [Mealchemy/nfr-tests/results/](#)